Attention to Sign: Multilingual Transfer Learning for Low-Resource Khmer Sign Language Recognition

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Abstract

Khmer Sign Language (KSL) is used daily by Cambodia's Deaf community, yet remains almost entirely unsupported by automatic recognition. Prior work from our institution found that MediaPipe keypoints generalize better than raw RGB frames under limited data. Building on this, we present a KSL isolated sign recognition system coupling an anatomically structured, visibility-aware Graph Convolutional Network (GCN) with a temporal Transformer encoder and a CLS token. Unlike a generic skeleton GCN, our spatial encoder excludes absent landmarks from message passing rather than zero-filling them, and masks a frame only when it carries no usable signal, so one-handed signs are never discarded. To address KSL's data scarcity, we adopt a two-stage cross-lingual strategy: a shared encoder is pretrained with per-language heads on six high-resource sign languages, then fine-tuned end-to-end, not frozen, on KSL with a new Khmer head, informed by evidence that unfrozen co-training transfers better than pretrain-and-freeze. On the 194-class KSL target dataset (combining a 100-class NISE corpus and a 94-class self-collected corpus), the full model attains a best validation accuracy of 93.58% (3.39M parameters) on a random per-class 80/20 split in the fully-supervised, cross-lingually co-trained setting, exceeding all recurrent baselines. As signers are shared between training and validation, this result is preliminary and does not measure generalization to unseen signers. By combining structural inductive bias with cross-lingual pretraining, this work aims to make practical, data-efficient sign recognition available for a still critically under-resourced language.

Keywords: *Khmer Sign Language Recognition, Graph Convolutional Network, Transformer, MediaPipe, Cross-Lingual Transfer Learning, Low-Resource Sign Language*

1 Introduction

More than 5.5% of the world's population lives with disabling hearing loss, projected to reach one in ten by 2050 [3]; in Cambodia, an estimated 1.5 million of ~16 million people are deaf or hard of hearing [1]. Khmer Sign Language (KSL) is this community's primary means of communication, yet has almost no digital infrastructure, and, being a full natural language with its own grammar, cannot be approximated by a resource built for another sign language or spoken Khmer [3]. Automatic KSL recognition could, in turn, underpin accessible tools such as sign-to-text captioning, educational dictionaries, and assistive communication technology, motivating the system this paper develops.

Two recent studies from our own institution, CADT, took the first steps here: Veng et al. [1] and Som et al. [2] each benchmarked RGB architectures against MediaPipe-keypoint models on 20-class KSL corpora (Section 2.2), converging on the same conclusion, skeletal landmarks generalize better than raw pixels under small-data conditions.

These studies are framed as preliminary baselines with three shared limitations that motivate this work: neither models the skeleton as a graph, so spatial relationships must be learned from scratch; neither treats missing landmarks explicitly, risking penalizing one-handed signs; and both train from scratch on a 20-class subset, leaving open whether pretraining on higher-resource sign languages could transfer useful structure to Khmer [4].

The rest of this paper is organized as follows. Section 2 reviews related work on isolated sign language recognition, prior KSL studies, pose- and graph-based recognition, and cross-lingual transfer. Section 3 details the proposed system: the mask-aware GCN over an anatomically

defined skeleton graph, the temporal Transformer with CLS-token pooling, and the two-stage cross-lingual pretraining strategy. Section 4 describes the experimental setup, including the datasets, pretraining sources, baselines, and evaluation protocol. Section 5 reports results, covering pretraining-language comparisons, a design ablation study, and few-shot evaluation. Section 6 discusses anticipated challenges and limitations, and Section 7 concludes with directions for future work.

The contributions of this work are:

- A new self-collected 94-class KSL video corpus of 2,045 clips from 6 signers, covering everyday vocabulary and recorded at varied speeds, angles, and conditions.
- A combined 194-class KSL target dataset of 5,052 clips, formed by merging this self-collected corpus with the existing 100-class NISE corpus, nearly a 10x expansion in vocabulary over the 20-class pilot corpora used in prior KSL studies [1], [2].
- A mask-aware GCN and temporal Transformer system with CLS-token pooling, cross-lingually pretrained on six source sign languages and co-trained end-to-end on KSL, that reaches a best validation accuracy of 93.58% on the 194-class target set (random per-class 80/20 split, signers shared across splits), exceeding a family of recurrent baselines (RNN, LSTM, BiLSTM, GRU, BiGRU); co-training was applied to the proposed model only.

2 Related Work

2.1 Isolated Sign Language Recognition

Isolated Sign Language Recognition (ISLR) maps a bounded clip containing a single sign to one class label, unlike continuous recognition or translation, which must segment and order multiple signs [5], [6]. Early systems paired CNNs with recurrent models; later work moved to 3D CNNs and Transformers capturing long-range dependencies via self-attention [1], [2], [6]. Being fixed-input and single-label, ISLR needs only a pooling or CLS-token head, not the autoregressive decoding translation requires [5].

2.2 Khmer Sign Language

KSL recognition is an emerging area with, at the time of writing, two published studies, both from CADT with the National Institute for Special Education (NISE), Phnom Penh. Veng et al. [1] benchmarked four RGB and three keypoint-sequence architectures on 20 KSL signs, finding SlowFast (RGB, 92.81%) and R3D-18 on keypoints (92.05%) strongest, though their non-signer-independent 70/30 split risks overestimating generalization. Som et al. [2] found a similar keypoint-over-RGB advantage (92.05% vs. 84.09%, improving to 94.32% with early stopping) and observed confusion between visually similar pairs like "Director" and "Female Director". We build on both findings, addressing the visually-similar-sign issue; a signer-independent split is left for future work.

2.3 Pose-Based and Graph-Based Sign Language Recognition

Pose-based recognition extracts structured keypoints via MediaPipe [9] or OpenPose, trading appearance for compact, background-invariant coordinates. GCNs go further, encoding skeleton connectivity as a graph, succeeding at spatial modeling in skeleton-based sign recognition [7]. TransSLR [4], targeting low-resource Central African Sign Language, uses a pure Transformer over normalized pose with no explicit graph, outperforming a multimodal RGB-pose baseline under signer-independent evaluation (RGB overfits to signer appearance). SignVLM [7] instead pairs a pretrained CLIP encoder with a Transformer decoder, transferring well in zero/few-shot regimes via CLIP's pretraining. Our approach sits between these: normalized pose geometry as in TransSLR, but with an explicit anatomical graph, suited to KSL where no CLIP-scale corpus exists.

2.4 Cross-Lingual and Multilingual Sign Language Recognition

Because most annotated sign language data is concentrated in a few high-resource languages, several works ask whether models transfer across sign languages despite distinct lexicons and grammars. Logos [8] introduces a large-scale RSL dataset and shows pretraining on it transfers to other-language ISLR, including few-

shot, but only for large corpora; critically, multi-dataset co-training outperforms both a frozen encoder and simple fine-tuning. TransSLR [4] offers a caution: a model pretrained on an ASL-scale corpus scores 0.00% zero-shot on CASL, "overconfident ignorance" from lexical divergence rather than a flawed representation. Together, these imply cross-lingual pretraining helps KSL only if the encoder stays trainable and the target language participates in shared co-training; Section 3.4 follows this evidence.

3 Methodology

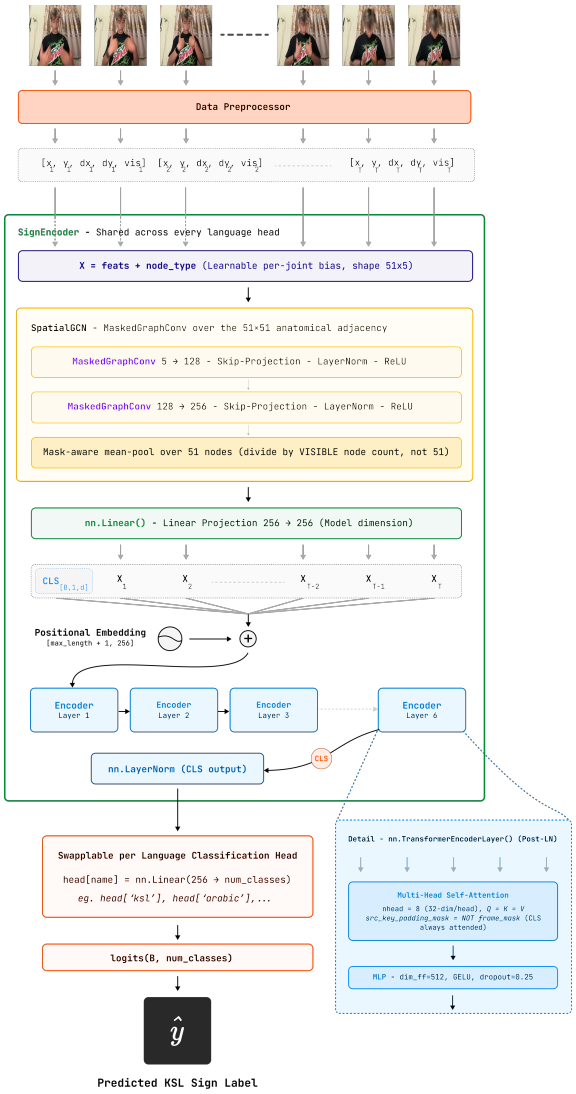


Figure 1. End-to-end system architecture, from raw video to predicted sign label.

Figure 1 summarizes the full pipeline, and each stage below builds directly on the previous one. A video clip is first converted into per-frame body and hand landmarks on a fixed anatomical skeleton (Section 3.1); these raw coordinates are normalized and paired with velocity and visibility channels (Section 3.2); a mask-aware spatial GCN then encodes each frame over that skeleton, explicitly excluding landmarks MediaPipe failed to detect rather than guessing their location (Section 3.3); the resulting per-frame embeddings pass to a temporal Transformer with a CLS token, the same encoder that carries the two-stage cross-lingual pretraining strategy (Section 3.4); and the augmentation suite and recurrent baselines used to train and evaluate every model close the section (Section 3.5). Full equations and symbol definitions are collected in Appendix A to keep this section focused on the design rationale.

3.1 Skeleton Representation and Anatomical Adjacency

Given a video clip of a single sign, MediaPipe Holistic extracts per frame a set of V=51 body and hand landmarks (9 upper-body pose points, 21 per hand; legs and feet dropped as noise), each with a coordinate and a presence flag. A fixed anatomical adjacency graph connects these nodes by skeleton structure rather than a learned or fully-connected topology, pose edges, a per-hand skeleton, and a wrist bridge linking each hand to the pose (Figure 2; block structure in Appendix B), reweighted by a learnable edge-importance matrix so the model can emphasize connections without inventing new ones.

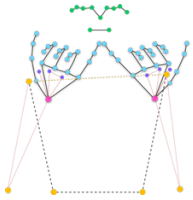


Figure 2. Anatomical skeleton graph: the GCN's fixed adjacency.

3.2 Feature Normalization

Each clip is normalized in two stages: body-frame centering on the shoulder midpoint and rescaling by shoulder width, then per-hand centering on each wrist, so handshape recognition is scale-invariant. Frame-to-frame velocity and a per-node visibility flag are appended as additional channels, kept separate from the normalized coordinates (Appendix A).

3.3 Mask-Aware Graph Convolution

A standard graph convolution assumes every node is present, which is false for sign-language landmarks: MediaPipe frequently fails to detect a hand that is out of frame, occluded, or resting during a one-handed sign. Our mask-aware convolution instead multiplies the adjacency by each frame's visibility vector before row-normalizing, so absent nodes contribute no influence rather than a fake location (full formulation in Appendix A); stacked layers are pooled by a visibility-weighted average into one feature vector per frame. A separate, coarser frame-level mask discards only frames with no detection at all, so one-handed signs are never dropped from training.

3.4 Temporal Transformer and Cross-Lingual Pretraining

Per-frame embeddings pass through a temporal Transformer encoder [17] with a prepended CLS token and learned positional encodings; a key-padding mask excludes undetected frames from attention, and the final CLS position becomes the clip embedding.

The shared encoder is pretrained jointly across six source sign languages, then fine-tuned end-to-end on KSL, never frozen, since freezing transfers markedly worse [8]; we compare this sequential protocol against a co-training variant that folds a KSL head into pretraining directly (Figure 3, Figure 4), avoiding the failure modes reported for naive cross-lingual transfer [4].

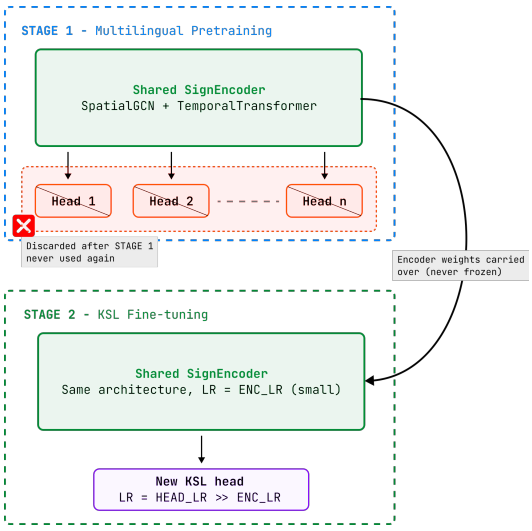


Figure 3. Incorporating KSL into cross-lingual training: sequential Stage 1-then-Stage 2

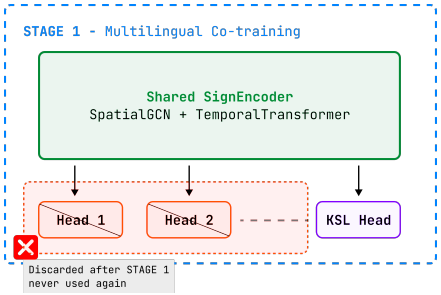


Figure 4. Incorporating KSL into cross-lingual training: co-training.

3.5 Augmentation and Recurrent Baselines

Augmentation perturbs the normalization anchors (anchor jitter, reach scaling) and applies temporal cropping, speed perturbation, frame dropping, and horizontal flipping, transforms tuned to survive normalization (Appendix A).

Five recurrent baselines, RNN [19], LSTM [20], BiLSTM [21], GRU [22], BiGRU [21], matching prior KSL work [1], are trained under an identical pipeline to isolate the graph-and-attention contribution.

4 Experimental Setup

4.1 Datasets

The target dataset combines two KSL sources: a 100-class corpus from the National Institute for Special Education (NISE), Phnom Penh, collected following our institution's earlier protocol [1], [2], and a 94-class corpus we collected ourselves from 6 signers on our own project team, covering everyday vocabulary across 194 signs (Table 1), recorded with digital cameras/smartphones at a standardized resolution and frame rate. Sign forms for the self-collected classes were taken from the public YouTube channel *Cambodian Sign Language For All* [23] and performed by our team members; the labels were not verified by Deaf signers or NISE teachers. This expands beyond the 20-class pilot used in [1], [2]. Classes are split by a per-class random 80/20 holdout (Section 4.5). Because both corpora mix several signers within each class and average only about 25 clips per class, a signer-independent split would leave too few clips per class for a reliable estimate and would require manual signer annotation of every clip. We therefore leave it to

future work, as [1] recommends. Figure 5 previews sample clips from the self-collected portion.



Figure 5. Sample frames from the self-collected KSL video corpus (six signers shown; faces blurred for privacy).

Pretraining datasets. Stage 1 (Section 3.4) draws on one large public dataset per source language: WLASL [10] for ASL (2,000+ glosses, 100+ signers); Slovo [11], among the largest public RSL isolated-sign datasets; and four further corpora covering Arabic [12], Argentine [13], Indian [14], [15], and Sri Lankan [16] Sign Language. All six are re-processed through the identical pipeline of Section 3.2.

Source languages were selected for public availability, a usable isolated-sign format, reliable annotation, a sufficient number of clips per class, and reasonable class balance. We excluded corpora that failed these criteria: British and German Sign Language datasets are continuous rather than isolated-sign, with complex annotation structure, and the Vietnamese Sign Language dataset has too few clips per class.

Table 1 makes the scale gap explicit: the 194-class KSL target set has only 5,052 samples, an order of magnitude fewer than Slovo (20,400) or Arabic (75,515), motivating the cross-lingual pretraining strategy. KSL's mean visibility (0.72) is comparable to Slovo (0.75) and Arabic (0.74), and well above the two lowest, WLASL (0.57) and Argentine (0.62), whose clips detect both hands less reliably.

Table 1. Dataset configuration for pretraining and fine-tuning.

Dataset	Role	Signs/Glosses	Signers	Samples	Mean Frames/Clip	Mean Visibility
KSL (NISE + Self Collected)	Target / fine-tune	194	6 + 6*	5,052	52	0.72
WLASL (ASL) [10]	Pretrain	2,000	119	12,505	67	0.57
Slovo (RSL) [11]	Pretrain	1,000	194	20,000	50	0.75
Arabic Sign Language [12]	Pretrain	502	[N/A]	75,515	40	0.74
Argentine Sign Language [13]	Pretrain	64	10	3,200	128	0.62
Indian Sign Language [14], [15]	Pretrain	192	[N/A]	2,999	65	0.81
Sri Lankan Sign Language [16]	Pretrain	383	[N/A]	4,236	32	0.62

Signs/Signers/Samples for public pretraining corpora are drawn from the cited sources;

* KSL, the self-collected portion used 6 signers (project team members) and NISE data were collected by 6 professional signers.

4.2 Exploratory Dataset Analysis

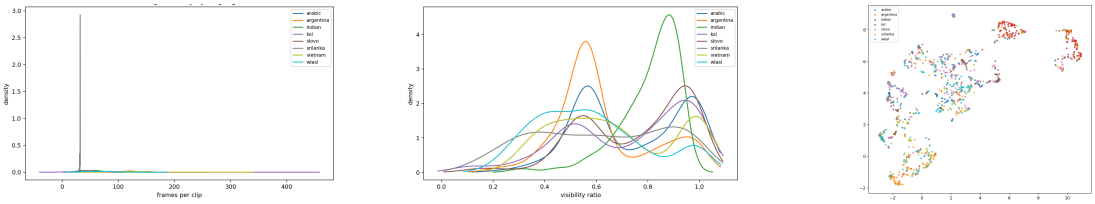
Before training, we ran the extraction and normalization pipeline of Sections 3.1-3.2 over 150 sampled clips per language to characterize the raw data (Figure 6).

Panel (a) plots per-clip frame-length density: most languages, KSL included, spread broadly past 150-200 frames, while Sri Lankan concentrates narrowly around 30 frames with a much taller peak, the spread the padding-plus-mask design (Sections 3.2, 3.4) is built to absorb.

Panel (b) plots per-clip visibility-ratio density (fraction of frames with both hands detected). Curves are bimodal but language-dependent: Indian and Slovo/Argentine skew toward a high-visibility peak (0.85-1.0), while KSL, Sri Lankan, and WLASL carry substantial density in a lower 0.4-0.6 band, supporting the node-level visibility masking of Section 3.3 rather than zero-filling missing hands.

Panel (c), a UMAP of a per-clip pose descriptor by language, shows substantial overlap across all seven datasets rather than

seven clean islands, motivating the multilingual pretraining strategy of Section 3.4: hand shape and motion already share structure across sign languages before any training.



(a) frame-length density *(b) visibility-ratio density* *(c) cross-lingual UMAP*

Figure 6. Cross-lingual exploratory statistics: (a) frame-length density; (b) visibility-ratio density; (c) cross-lingual UMAP.

Figure 7 focuses on the KSL target dataset. Panel (a), the clip-length histogram over 500 clips, is strongly right-skewed, peaking at 40-80 frames with a thin tail past 150, reinforcing the need for masked, variable-length batching.

Panel (b), a UMAP of the top 15 KSL classes, shows partial separability, some classes cluster tightly, others stay mixed, indicating raw pose statistics carry signal but leave real representational work for the GCN and Transformer.



(a) KSL clip-length distribution (n=500 sampled) *(b) KSL per-class UMAP (top 15 classes)*

Figure 7. KSL target-dataset statistics: (a) clip-length distribution; (b) per-class UMAP.

4.3 Preprocessing Pipeline

Every clip is processed by the same pipeline: MediaPipe Holistic extracts per-frame pose and hand landmarks, trimmed and combined into the *V=51* layout (Section 3.1); the dual-stage normalization and velocity/visibility assembly of Section 3.2 is applied per clip; and clips within a batch are zero-padded to the batch's longest sequence, with an accompanying visibility tensor and frame-usability mask so padding is never mistaken for a real frame. Clips shorter than a minimum frame threshold are dropped, since one or two frames almost always indicates a failed extraction.

4.4 Training Protocol

Stage 1 (multilingual pretraining) trains the shared encoder and the six source-language heads jointly with a per-language batch composition, the default augmentation preset (Section 3.5), and an AdamW optimizer with weight decay [18], consistent with recent sign language recognition work [7], [8]. The learning rate follows a linear warm-up into a plateau schedule that halves it on a validation-accuracy stall, gradients are clipped to a fixed norm, and the loss is label-smoothed cross-entropy (plain cross-entropy only for reported validation loss); the CLS token, positional encodings, and node-type embeddings are initialized from a truncated normal distribution, other layers at PyTorch defaults. Stage 2 (KSL fine-tuning) discards the source-language heads, attaches a new Khmer head, and fine-tunes end-to-end with a smaller encoder learning rate and the lighter augmentation preset. The recurrent baselines (Section 3.5) use the identical two-stage protocol and hyperparameters, changing only the encoder, for a fair comparison.

4.5 Evaluation Protocol

Evaluation uses a random per-class 80/20 train/validation split (Section 4.1). No separate

test set is held out, and all reported accuracies are validation accuracies, with the best configuration selected on the same split. Signers appear in both splits, although clips within a class vary in recording speed, angle, and conditions. Results are therefore preliminary and do not establish signer-independent generalization , which [1] recommends, is left to future work. We report two regimes for the KSL target: a few-shot setting, fine-tuning the Khmer head (and, optionally, the encoder) on K = 1, 3, and 10 samples per class; and a fully-supervised setting, fine-tuning on the entire training split, the number most comparable to [1] and [2]. For few-shot evaluation, each model is initialized from the Stage-1 language subset on which it performed best (Table 3). We report top-1 accuracy for every setting.

5 Results

This section reports the complete experimental results for the system of Sections 3-4, following the two-stage protocol of Section 4.4. Table 2 lists Stage-1 pretraining accuracy across six source-language combinations, isolating how much the choice of pretraining languages matters. Table 3 lists Stage-2 KSL accuracy under those combinations (plus a from-scratch control), reporting sequential fine-tuning and multi-dataset co-training (Section 3.4) side by side. Table 4 lists the ablation study, and Table 5 reports the few-shot results of Section 4.5.

In Stage-1 pretraining (Table 2), the proposed model outperforms all five recurrent baselines in every language combination, and the margin widens as the combination shrinks: on the hardest two-language split (RSL & WLASL) it reaches 46.25% against BiGRU's 40.02%, and on WLASL alone 31.13% against 26.68%, while on the full six-language set the gap narrows to about one point (76.78% vs. 75.65%), suggesting the architecture's advantage is largest precisely where pretraining data is scarcest. In Stage-2 fine-tuning (Table 3), the sequentially pretrained model is competitive with the baselines, with its largest gain for Arabic + Argentina + RSL (92.23% vs. 90.72% for BiGRU).

The best overall validation accuracy, 93.58% (3.39M parameters), comes from co-training with the Arabic + Argentina + RSL subset, ahead of the same subset's sequential-pretraining result (92.23%) and every recurrent baseline's best score (BiGRU’s 91.90%, trained from scratch). Trained from scratch on KSL alone, however, the proposed model (84.90%) trails BiGRU (91.90%): the recurrent baseline is the stronger small-data learner without pretraining, so the architecture's benefit is concentrated in how well it exploits cross-lingual signal rather than in raw from-scratch capacity.

Three ablations are completed. Removing the frame-usability rule (masking any frame missing a hand, rather than only fully-undetected frames) causes by far the largest drop, 29.31 points (93.58% to 64.27%), since many KSL signs are one-handed and this ablation discards the very frames carrying their meaning. Removing node-level visibility masking (zero-filling absent joints instead of excluding them) costs 5.33 points (88.25%), and replacing the anatomical adjacency with a fully-connected graph costs 4.54 points (89.04%); both results support the mask-aware, structurally-grounded design of Section 3.3.

Table 2. Stage-1 multilingual pretraining accuracy by language combination.

Model	All	Exclude WLASL & Srilanka	Arabic + Argentina + RSL	Exclude RSL & WLASL	RSL & WLASL	RSL	WLASL
RNN	65.62	73.33	73.30	77.23	26.12	31.12	18.05
LSTM	75.10	74.11	81.33	85.14	37.02	45.49	21.45
BiLSTM	75.15	89.64	82.56	85.28	37.69	47.09	23.17
GRU	74.67	74.66	81.67	84.55	37.18	47.89	22.45
BiGRU	75.65	89.77	83.25	85.32	40.02	52.35	26.68
Mask-Aware GCN + Transformer (ours)	**76.78**	**89.87**	**87.84**	**86.44**	**46.25**	**68.38**	**31.13**

Each column pretrains the encoder on a different language subset (Section 4.1); values are mean per-language validation accuracy.

Table 3. Stage-2 KSL fine-tuning validation accuracy by pretrained-language combination.

Model	From scratch	All	Exclude WLASL & Srilanka	Arabic + Argentina + RSL	Exclude RSL & WLASL	RSL & WLASL	RSL	WLASL
RNN	81.74	86.33	84.06	88.28	83.17	85.21	86.15	83.75
LSTM	89.84	88.71	89.27	89.13	88.54	89.48	90.21	88.54
BiLSTM	90.09	89.65	89.90	89.94	89.58	90.31	90.73	89.58
GRU	85.84	88.45	89.55	89.53	88.12	90.62	89.69	86.46
BiGRU	**91.90**	89.60	89.92	90.72	90.42	90.83	91.56	89.79
Ours (pretrained)	84.90	89.68	**90.05**	92.23	89.95	90.82	**91.58**	**91.25**
Ours (co-training)	84.90	**91.13**	89.33	**93.58**	**90.72**	**90.89**	90.08	90.03

"From scratch" trains Stage 2 directly on KSL with no pretraining; other columns fine-tune from the corresponding Stage-1 encoder (Table 2). Co-training (Section 3.4) was evaluated only for the proposed model; baselines are reported under sequential pretraining only.

Table 4. Ablation study on KSL (fully-supervised setting).

Configuration	Validation Accuracy	Δ vs. full model
Full model (mask-aware GCN + Transformer + multilingual pretrain)	**93.58 %**	—
– node-level visibility masking (zero-fill absent joints instead)	88.25 %	-5.33 pp
– frame-usability rule (mask any frame missing a hand)	64.27 %	-29.31 pp
– anatomical adjacency (fully-connected graph instead)	89.04%	-4.54 pp

Each row removes exactly one design element from the full model, isolating its individual contribution.

Table 5 reports the few-shot (K = 1, 3, 10) results of Section 4.5; visually similar KSL sign pairs (e.g., "Director" vs. "Female Director") [2] are discussed further in Section 6.[2]

Table 5. Few-shot (K = 1, 3, 10) top-1 accuracy (%). Each model uses its own best-performing Stage-1 encoder, selected per model.

Model	K = 1	K = 3	K = 10
RNN	36.51	56.69	73.29
LSTM	37.03	57.34	79.36
BiLSTM	43.92	63.67	82.81
GRU	39.17	59.12	82.11
BiGRU	44.28	65.13	82.89
Mask-Aware GCN + Transformer (ours)	**45.82**	**67.95**	**83.93**

Each K column fine-tunes the Khmer head (and, optionally, the encoder) on K labeled samples per class (Section 4.5). These results were obtained over 5 trials and 60 epochs.

6 Discussion, Anticipated Challenges, and Limitations

Evaluation protocol. Results are validation accuracies on a random per-class 80/20 split in which signers are shared between training and validation, and the best configuration, including the few-shot encoder for each model, was selected on the same split. Per-class sample counts (about 25) were too small to support a separate test set, and signer labels are not available for every clip. The reported accuracies may therefore overestimate performance on unseen signers and should be read as preliminary. Baselines were not co-trained, so the co-training gain is not directly comparable. Signer-independent train/validation/test evaluation is the priority for future work.

Visually similar signs. Both prior KSL studies report confusion between semantically similar pairs [1], [2], the visually similar signs (VSS) problem [8]. Our model excludes facial expression, so purely non-manual distinctions are invisible to it; facial-landmark support is the natural fix.

Cross-lingual transfer risk. Naive transfer can fail outright when languages diverge [4]. KSL's relationship to the six pretraining languages is uncharacterized; K=1 few-shot results (Section 4.5) measure this risk, and the unfrozen co-trained strategy (Section 3.4) degrades gracefully rather than catastrophically.

Dataset scale. Logos [8] found corpus size critical to transfer quality; our Stage-1 sources are among the largest available, but KSL

remains small by comparison, a gap the few-shot protocol (Section 4.5) characterizes honestly.

Recording protocol inconsistency. The self-collected KSL clips (Section 4.1) predate a consistent framing protocol, so the signer's face or hips sometimes fall outside frame, unlike NISE's. Since normalization anchors on visible pose landmarks (Section 3.2), this can read as a real visibility pattern rather than a framing artifact.

Signer and environmental diversity. Following both prior studies [1], [2], the target dataset uses more signers and varied conditions than earlier 20-class pilots; but signers are shared across splits and the NISE signer count is not yet annotated. The self-collected signs were learned from public videos [23] and performed by non-fluent team members without expert verification, so some clips may deviate from standard KSL articulation. Expert verification is left to future work.

Real-time inference speed. Informal testing on a separate web deployment (GCN+Transformer, BiLSTM, BiGRU) found the Transformer fastest overall and BiGRU a close second. Not part of this paper's formal benchmarks; code at https://github.com/Phal-Sovandy/Khmer-Sign-Language-Translation-System

7 Conclusion and Future Work

This paper presented a KSL isolated recognition system built around a mask-aware GCN, a temporal Transformer, and a two-stage cross-lingual pretraining strategy. It addresses two limitations of prior KSL work [1], [2], anatomical structure and missing-landmark handling, leaving signer-independent evaluation as future work; the reported accuracies are validation results and should be considered preliminary, full results are in Section 5. Future work includes facial landmarks, a larger vocabulary and signer pool, expert verification of sign labels, signer-independent evaluation, model compression for lighter-weight deployment, and continuous recognition and translation. Code: https://github.com/Phal-Sovandy/ksl-recognition

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Appendix A. Mask-Aware Graph Convolution

This appendix gives the full mask-aware graph convolution introduced in Section 3.3, together with the definition of every term.

$$H_t = D_t^{-1}\left((A \odot M + I) \odot v_t\right)(X_t\, W) \quad \text{(A1)}$$

where H_t is the layer's output node features for frame t, A is the fixed anatomical adjacency, M the learnable edge-importance weights, I the self-loop identity matrix, v_t the frame's node-visibility vector, D_t the resulting row-sum (visible-degree) diagonal matrix, X_t the input node features, and W the layer's learned weight matrix.

Appendix B. Anatomical Adjacency Structure

This appendix gives the block structure of the fixed anatomical adjacency matrix A introduced in Section 3.1. The $V = 51$ nodes are ordered as 9 upper-body pose nodes, 21 left-hand nodes, and 21 right-hand nodes. The adjacency A is a 51 × 51 matrix that decomposes into a block matrix over the pose, left-hand, and right-hand node groups:

$$A = \begin{bmatrix} A_{pose} & B_L{}^T & B_R{}^T \\ B_L & A_{lhand} & 0 \\ B_R & 0 & A_{rhand} \end{bmatrix} \quad \text{(B1)}$$

where A_{pose} (9x9), A_{lhand} and A_{rhand} (21x21 each) are the within-group bone connections, and B_L, B_R (21 × 9) are all-zero except for one entry, the bridge edge linking that hand’s wrist to the pose wrist. The hands therefore never connect directly, only through the pose block. The learnable edge-importance matrix M in (A1) shares this same block-sparsity pattern, so it can reweight existing edges but cannot create new ones.